

# **NYC IBX Commute Relief Estimator**

Predicting the commute impact of the Interborough Express using machine learning

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Course: DATA 622 Machine Learning Project

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Spring 2026

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# NYC IBX Commute Relief Estimator

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## 1. Problem Definition

Outside of Manhattan, New York's boroughs have some of the longest commute times in the country. According to the Pratt Center for Community Development, average commutes in some outer borough neighborhoods run as high as 53 minutes (Pratt Center, 2014). Of the ten neighborhoods with the longest commutes in the city, eight are in Brooklyn or Queens. A big part of the reason is geography: the subway system was built to transport people in and out of Manhattan, not to move them across Brooklyn and Queens. There is no fast crosstown connection between the two boroughs.

The MTA's proposed Interborough Express is meant to change that. The IBX would run along an existing 14-mile freight rail corridor from Bay Ridge, Brooklyn to Jackson Heights, Queens, with 19 stations serving neighborhoods that the MTA describes as historically underserved by rapid transit. According to the MTA, close to 900,000 residents live along the proposed route, nearly three-quarters of whom are people of color and 19% of whom live below the poverty line. The project entered engineering and design in 2025 and is currently in active environmental review (MTA, 2025).

**The question this project tries to answer:** What will the IBX's neighborhood-level impact look like, before the line is built? Specifically, will there be relief for workers with long (60-minute-plus) commute times?

Traditional before-and-after studies require waiting years for infrastructure to open. This project builds a what-if tool instead: train predictive models on existing census, transit, and mobility data across all of New York City, where we already know how transit access and commute patterns relate to each other, then simulate what happens when you add transit access to the IBX corridor.

The result is a census-tract-level sensitivity analysis. Improve subway access in the treated tracts and see how the predicted share of 60-minute-plus commuters moves. That difference is the estimated commute relief for each neighborhood.

Our target users are NYC government personnel, transit planners, and curious New Yorkers. The app is built so a non-technical user can explore projected impact by neighborhood, look at the demographic context behind the predictions, and compare how different modeling approaches forecast the same proposed change.

## 2. System Design

The whole project runs in R, from data pull to modeling to the Shiny app. Knitting the R Markdown file regenerates every dataset and model object. The app reads those saved outputs and serves them interactively.

## 2.1 Data Sources

We pulled from three main sources:

- 2022 ACS 5-Year Estimates. Census-tract-level data for all five boroughs: income, education, commute mode, work-from-home rates, population density, poverty, housing, car ownership, and age. We picked 2022 because it is post-pandemic reopening and has the stability of a 5-year rolling average.
- TLC Trip Records, October 2023. Uber/Lyft and taxi trip counts from the NYC Taxi and Limousine Commission. October avoids the holiday season while staying close to the 2022 census window. We aggregate by pickup taxi zone and then weight into census tracts by area overlap.
- Citi Bike Trip Data, October 2023. Station-level trip counts for the same month. We geocode each station to its census tract to get tract-level bike trip volume and station density.

One important additional source we used specifically for the what-if analysis is:

- MTA IBX SEQRA Draft Scoping Document, 2025: The MTA's Draft Scoping Document for the Interborough Express, released as part of the state environmental review process (SEQRA). We extracted the addresses of all 19 proposed IBX station locations from this document, geocoded them to coordinates, and used them as the basis for the IBX simulation.

We also computed two spatial features ourselves:

The first is distance from each tract to the nearest subway entrance, using the 2024 MTA entrance dataset. We use `st_point_on_surface()` instead of `st_centroid()` here because some tracts are L-shaped or coastal, and their centroids land in water. The second is distance to Grand Central Terminal as a proxy for proximity to Manhattan.

## 2.2 Data Preparation

The trickiest part of data preparation is that census tracts and taxi zones are completely different geographies which do not line up. We handle this by projecting both into EPSG:2263 (a feet-based coordinate system built for the NY/Long Island area), running `st_intersection()` to find all the places where tracts and zones overlap, and then distributing taxi trips across tracts proportionally by area. Thinking of it like a Venn diagram, trips get divided up based on how much each census tract overlaps with the taxi zone they came from.

On the census side, we engineer percentage features from raw counts: `pct_commute_60p` is our target (workers with 60+ minute commutes divided by total workers), and we also build per-capita trip rates and station density features. The four most skewed predictors (uber trips, bike trips, station density, and population density) get a `log1p` transform before z-scoring. Tracts with fewer than 50 residents or missing transit data are dropped. Missing income and age values are imputed at the borough level using the local median.

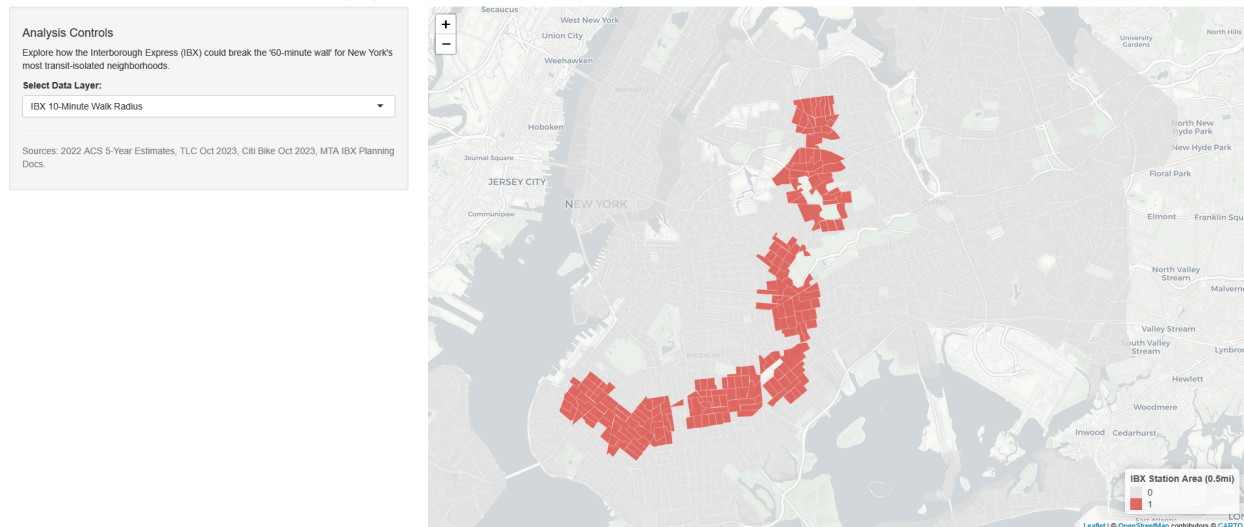
## 2.3 IBX Simulation

For the IBX, we geocoded all 19 proposed station addresses from the MTA's 2025 Environmental Scoping Document. We draw a 0.5-mile buffer around each station (about a 10-minute walk, which is the standard threshold for transit planning), then flag every census tract that intersects with any of those buffers as treated.

The simulation itself is simple: for treated tracts, we swap `dist_subway_miles` to the best value in the dataset, which is the closest any tract in NYC sits to a subway entrance. Then we run predictions through

each model. The difference between the baseline prediction and the IBX prediction is the estimated commute relief in percentage points. It is a simplification, but a reasonable one for this stage of planning.

The 60-Minute Wall: NYC Commute Equity & IBX Simulator



**Figure 1. MAP OF 19 IBX STATION WALKSHEDS OVERLAID ON NYC CENSUS TRACTS.**  
**ibx\_geocoded map from the ibx-simulator-data chunk**

## 2.4 Stack

R throughout. Key packages: `tidycensus`, `sf`, `tigris`, `spdep` for spatial work; `arrow` for the TLC Parquet files; `randomForest` and `splines` for modeling; `rsample` for cross-validation; and `Shiny` for the app.

## 3. Machine Learning Components

We built three types of predictive models (Linear, Linear + Spline, Random Forest), each tested with and without spatial-lag features, for six predictive model variants total. Plus we built one causal model (which did include the spatial-lag features).

The predictive model types are a linear regression baseline, a linear model with a natural cubic spline on subway distance, and a Random Forest. Each takes the same 23 features (with or without the spatial-lag features depending on the variant) and predicts `pct_commute_60p`. Running all three lets us directly compare how model choice affects the IBX simulation results, which ultimately mattered a lot. We also developed a BART (Bayesian Additive Regression Trees) model for causal inference, and we included the spatial-lag features.

### 3.1 Features

All three predictive model types share the same 23 features, grouped below:

Borough fixed effects (Brooklyn as reference), distance to nearest subway entrance, distance to Grand Central as a Manhattan proximity proxy, Uber/taxi trips per capita, Citi Bike trips per capita, Citi Bike stations per 1,000 residents, stations per square mile, median income, poverty rate, vacancy rate, renter share, share of households with 2 or more cars, population density, share with no car, share commuting by transit, share working from home, share walking to work, share with a bachelor's degree, and five

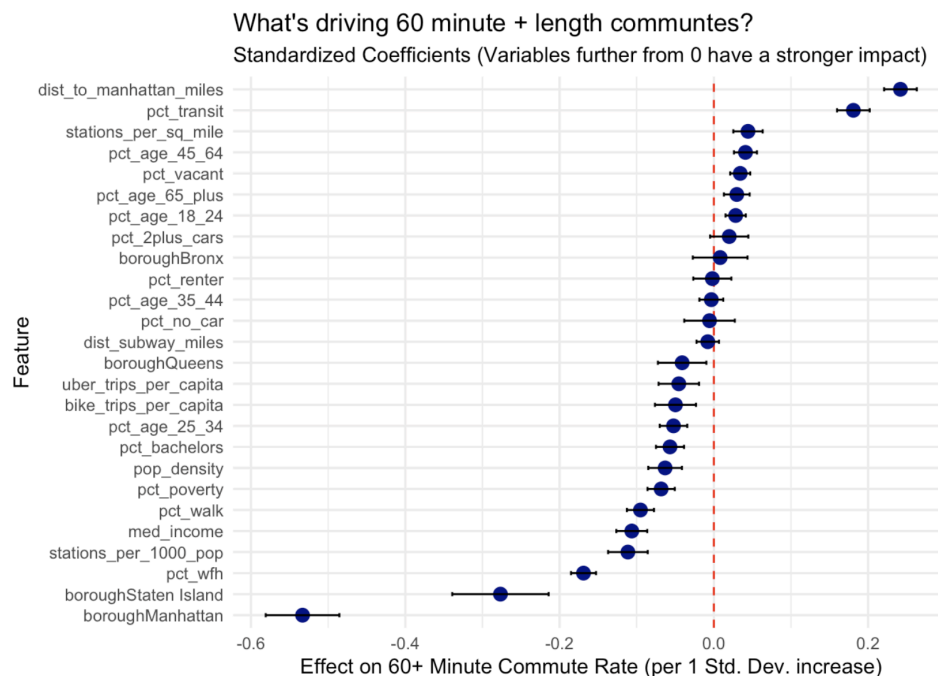
age-bracket shares (18-24, 25-34, 35-44, 45-64, 65+). We also added spatial-lag versions of seven geographic predictors (distance to subway, distance to Manhattan, transit share, WFH share, Citi Bike stations per capita, income, and population density). For a given tract, the spatial-lag features were calculated using the weighted average for each of those features among the neighboring tracts. Each of the three predictive model types was fit twice, once with these lag features and once without, totaling six models. The model variants performed very similarly to one another when they had the same base model type, so for the sections below, we will speak to each model type we tried rather than repeat the same findings for both the spatial-lag and non-spatial-lag variants of the same model type.

### 3.2 Linear Regression

The linear model type regresses `pct_commute_60p` on all 23 features. We logit-transform the outcome before fitting, so predictions stay in the valid 0 to 100 range and back-transform at prediction time.

The most counterintuitive result is that the linear model finds `pct_transit` has a positive coefficient: tracts where more workers commute by transit have more 60-minute-plus commutes, not fewer. This is likely a geographic effect rather than a causal one. Tracts with high transit use tend to be in the outer boroughs, far from job centers, where commutes are long regardless of mode.

The coefficient on `dist_subway_miles` is essentially zero in the pure linear model type, which is why it predicts almost no relief from the IBX. The issue is the saturating subway-access effect: a 5-minute versus 10-minute walk to the subway is a very different commute experience for most New Yorkers, but a 20-minute versus 25-minute walk is not. A linear term cannot capture that distinction, which is the motivation for Model Type 2.



**Figure 2. STANDARDIZED COEFFICIENT PLOT.**  
model-outputs chunk, lm predictors sorted by effect size

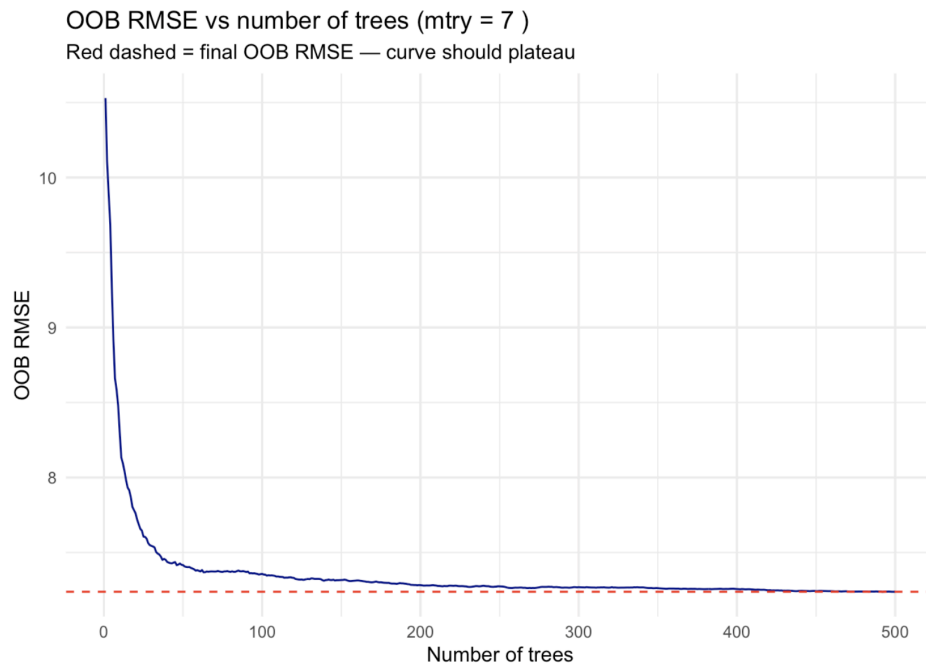
### 3.3 Linear + Natural Cubic Spline

Model Type 2 is a linear model that incorporates a natural cubic spline. It addresses the saturating subway-access effect described earlier by replacing the linear `dist_subway_miles` term with a natural cubic spline ( $df = 4$ ). The spline learns different slopes for different ranges of subway distance, so it can capture the steep drop-off in benefit once you get past a certain walking distance.

This has a big effect on the IBX simulation. The linear model predicts close to zero relief because it cannot fully capture what a new nearby station does for a neighborhood. The spline model predicts around 7 to 8 percentage points of relief for treated tracts, which is a much more plausible estimate.

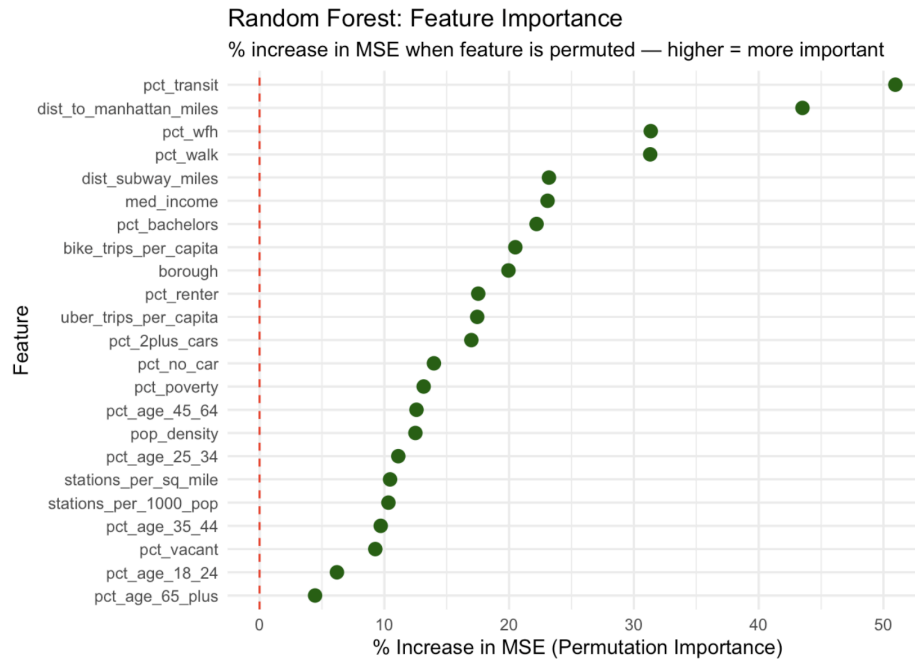
### 3.4 Random Forest

The Random Forest model type uses 500 trees and `mtry = 7` (the standard  $p/3$  default). We did a sweep over `mtry` values of 3, 5, 7, 11, and 22 using out-of-bag error and all five landed within 1% of each other, so we stuck with the default rather than reporting a marginal algorithmic winner. The OOB error curve flattened well before 500 trees.



**Figure 3. RF TUNING PLOT.**  
OOB RMSE vs Number of trees

RF's top predictors largely agree with the linear model (`pct_transit`, `dist_to_manhattan_miles`, and `pct_wfh`), but RF ranks `dist_subway_miles` much higher, which is consistent with the saturating subway-access effect described in section 3.2. RF likely also captures interactions between features that the linear models cannot, which is a natural advantage of tree-based methods.



**Figure 4. RF FEATURE IMPORTANCE PLOT**

rf-feature-importance chunk, % increase in MSE when each feature is permuted

### 3.5 Cross-Validation and Model Comparison

All six predictive model variants (Linear, Linear + Spline, and Random Forest, each with and without spatial-lag features) go through the same 10-fold cross-validation setup, stratified by borough. We reuse the same folds across variants, so any differences in performance come from the model, not the data split. We use the one-SE rule to flag the simplest model within one standard error of the best.

| Model                               | RMSE_mean | RMSE_SE | R2_mean | R2_SE | within_1se_of_best |
|-------------------------------------|-----------|---------|---------|-------|--------------------|
| Linear Regression                   | 7.768     | 0.118   | 0.610   | 0.011 |                    |
| Linear Regression + Spatial Lags    | 7.623     | 0.118   | 0.624   | 0.011 |                    |
| Linear + Spline (dist_subway_miles) | 7.606     | 0.135   | 0.626   | 0.013 |                    |
| Linear + Spline + Spatial Lags      | 7.496     | 0.136   | 0.637   | 0.013 |                    |
| Random Forest                       | 7.236     | 0.140   | 0.661   | 0.013 | ✓                  |
| Random Forest + Spatial Lags        | 7.138     | 0.144   | 0.670   | 0.013 | ✓                  |

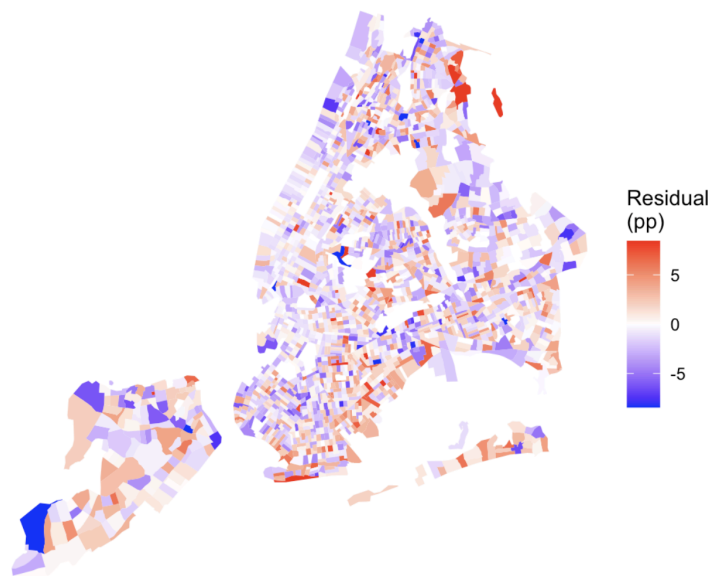
**Table 1. 10-FOLD CV RESULTS TABLE**

random-forest-model chunk, cv\_table output

Random Forest + Spatial Lags wins at 7.14 RMSE, with plain Random Forest (RF) essentially tied within one standard error at 7.24. Two patterns stand out in the table. First, every model class improves with spatial-lag features, by a consistent 0.1 to 0.15 RMSE, which tells us neighbors carry real signal beyond what a tract's own features capture. Second, the spline-to-RF jump is bigger than the linear-to-spline jump, which means Random Forest is doing something the spline alone cannot, likely capturing interactions between features.

While developing the models, we also checked whether the model's prediction errors clustered geographically, which would suggest it was missing something spatial. Before adding `dist_to_manhattan_miles`, they did cluster. After adding it, that clustering dropped by roughly half, and RMSE improved across all three model types, suggesting Manhattan proximity was a significant geographic gap the model had been missing.

RF residuals — actual minus predicted % long commute  
Red = model under-predicts | Blue = model over-predicts



**Figure 5. RF RESIDUAL MAP**

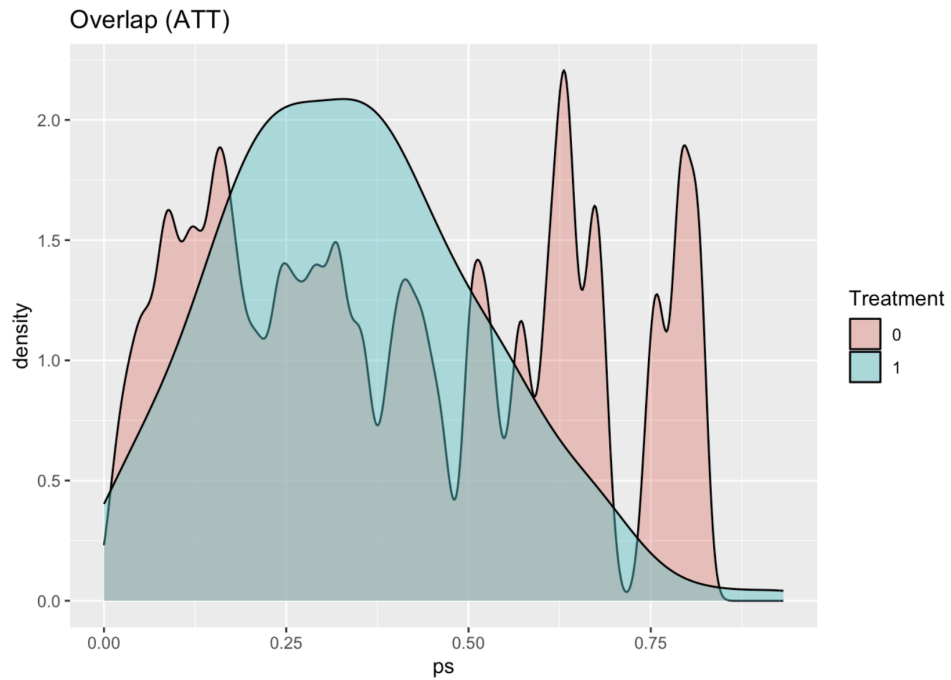
rf-residual-diagnostic chunk, actual minus predicted % long commute

### 3.6 BART: Bayesian Causal Inference

The three predictive model types estimate commute relief based on observed associations in the data. BART goes a step further and attempts a causal estimate: given that a tract received IBX access, how much did that actually reduce its share of 60-minute-plus commuters compared to the counterfactual where it did not?

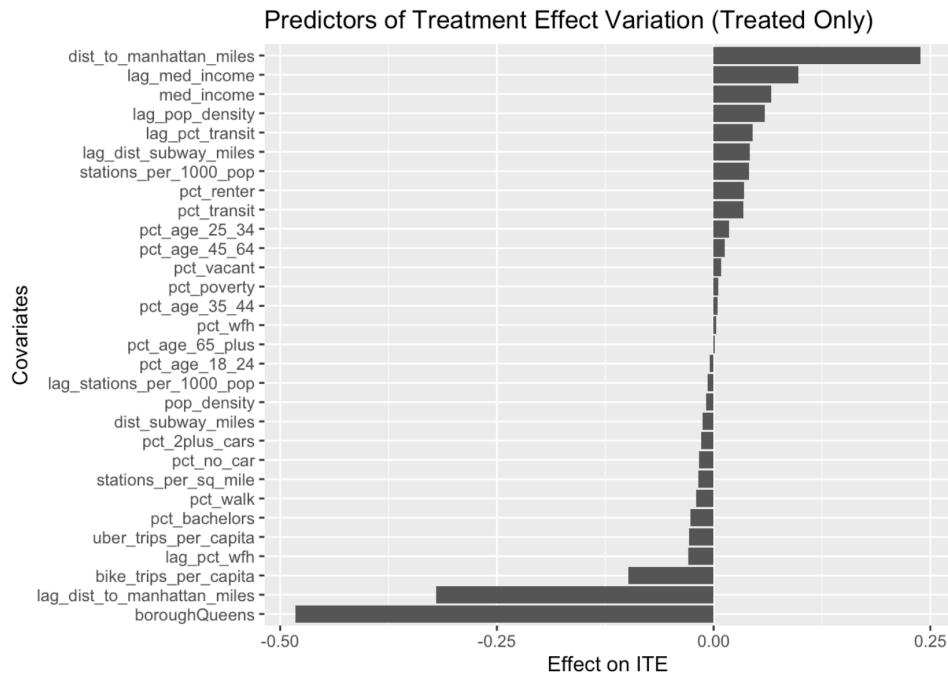
For this causal model, we use BART to estimate the Average Treatment Effect (ATE): on average across all tracts, how much does the IBX intervention change the share of 60-minute-plus commuters? We then report the ATE values for treated tracts specifically (Queens, Brooklyn), which is where the policy question lives.

We calculated propensity score overlap between our treated tracts (on IBX corridor) and not treated tracts (outside the corridor). Trustworthy causal estimates require good overlap, where the two distributions both span the same range. Most of our tracts have similar-looking untreated tracts to compare each treated tract against. There are a few areas with poor overlap (e.g., the red spikes at high propensity values) meaning the BART model has to extrapolate for them, comparing treated tracts to controls that don't really resemble them.



**Figure 6. BART ATT PROPENSITY OVERLAP PLOT**  
Some tracts have no good comparison group

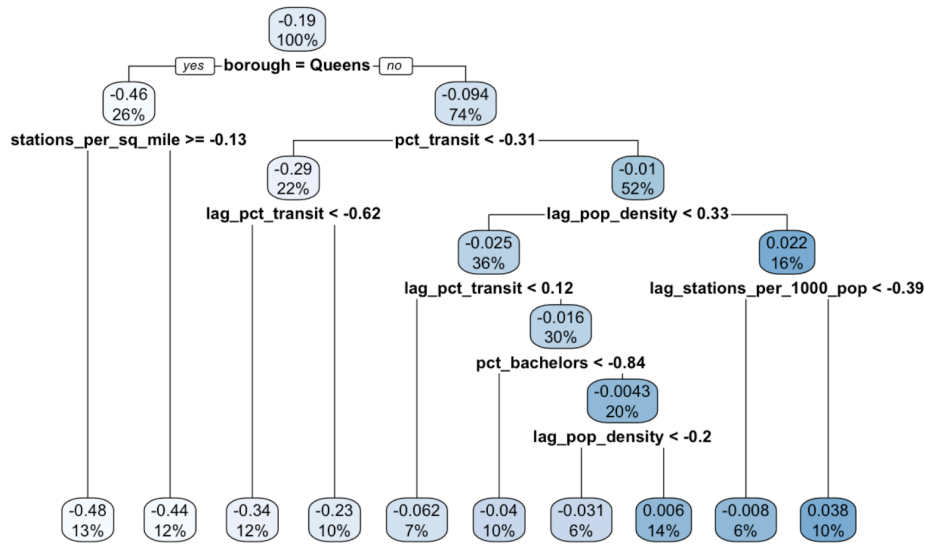
Expect shorter commutes if you're in a treated tract in Queens, with neighboring tracts far from Manhattan, with lots of CitiBike commuters, and not that many train stations, but surrounded by other tracts with high station density.



**Figure 7. BART**  
Predictors sorted by Effect on ITE

Among the treated tracts, we were curious about which subgroups would benefit most. We created a BART Regression Tree on ITE Plot for treated tracts only. Each split is the question that best separates "tracts with bigger benefit" from "tracts with smaller benefit." Each terminal node shows the predicted ITE for tracts that follow that path (the % at the bottom shows what share of tracts ended up there). Looking at the demographic profile of who benefits most, they are more likely to: be in Queens, be further from the subway, be currently less transit-dependent, and be less population dense areas.

### Subgroups Driving Treatment Effect Heterogeneity



**Figure 8. BART REGRESSION TREE ON ITE PLOT**  
For treated tracts only

## 4. Application Demonstration

The Shiny app is the main deliverable for end users. It is built for transit planners and policy staff who want to explore IBX impact by neighborhood without touching any R code.

### 4.1 IBX Predictive Simulation

Users can switch between all six model variants and see a choropleth map of estimated commute relief in percentage points across NYC census tracts, with IBX-treated tracts highlighted. Relief is labeled as Drop in Long-Commute Share (percentage points), reflecting that the model predicts the share of workers with 60-minute-plus commutes, not individual commute times. Percentage points here represent the difference between baseline and IBX predicted commute shares.

The difference between models is visible and worth showing to users. The linear model predicts near zero relief everywhere. The spline and RF models both show a concentrated band of relief running through the IBX corridor. The spatial-lag variants extend that band to roughly 113 neighboring tracts as well, since the lag features let predicted relief spill over to tracts adjacent to the corridor even when they are not directly within walking distance of an IBX station. That contrast is itself part of the story: a linear model will miss the IBX's impact almost entirely if the relationship between transit access and commute outcomes is nonlinear.

## The 60-Minute Wall: NYC Commute Equity & IBX Simulator

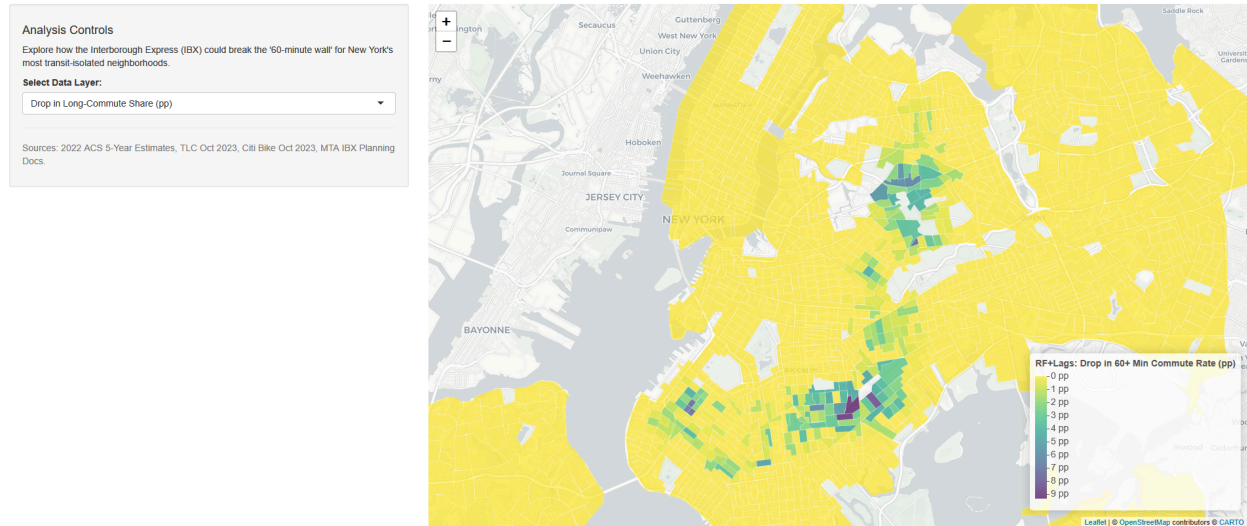


Figure 9. SHINY APP: IBX SIMULATION PANEL, RF MODEL SELECTED

### 4.2 Demographic and Infrastructure Layers

Additional layers let users explore the context behind the predictions: median income, poverty rate, population density, distance to Manhattan, and bike infrastructure density. The idea is to let someone ask not just "where does IBX help?" but "who does IBX help?".

### 4.3 RF Residuals Map

There is also a residuals layer that shows actual minus predicted long-commute share, capped at the 99th percentile for readability. Red tracts are places where the model underpredicts, meaning real long-commute rates are higher than expected. These tracts may point to a missing feature the model cannot see, like specific bus network gaps or employment geography. It is mainly a diagnostic layer, but we surface it in the app for transparency.

The 60-Minute Wall: NYC Commute Equity & IBX Simulator

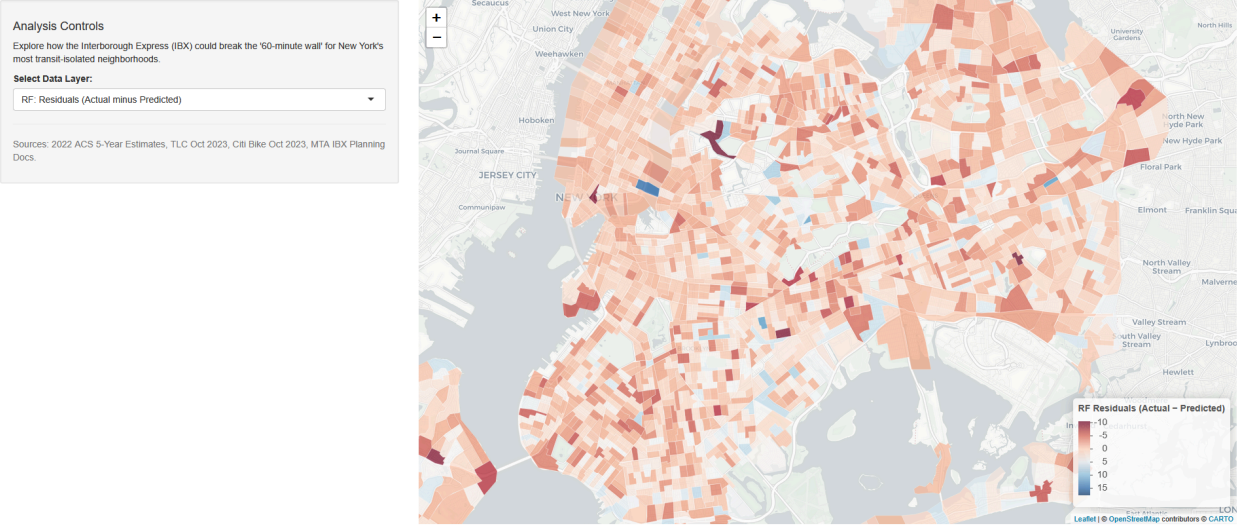


Figure 10. SHINY APP: RF RESIDUALS LAYER, DIVERGING RED/WHITE/BLUE COLOR SCALE

4.4 BART Treatment Effect Map

BART agrees with the predictive models simulation in direction, and that the IBX is reducing long commutes. Queens treated tracts benefit most, with an average effect around 0.46 percentage points of reduction. Brooklyn benefits too, but less, around 0.09 pp on average. The BART magnitudes are smaller than the predictive simulation's because BART estimates an average causal effect with Bayesian shrinkage toward zero, while the predictive simulation reports max relief under an extreme scenario.

The 60-Minute Wall: NYC Commute Equity & IBX Simulator

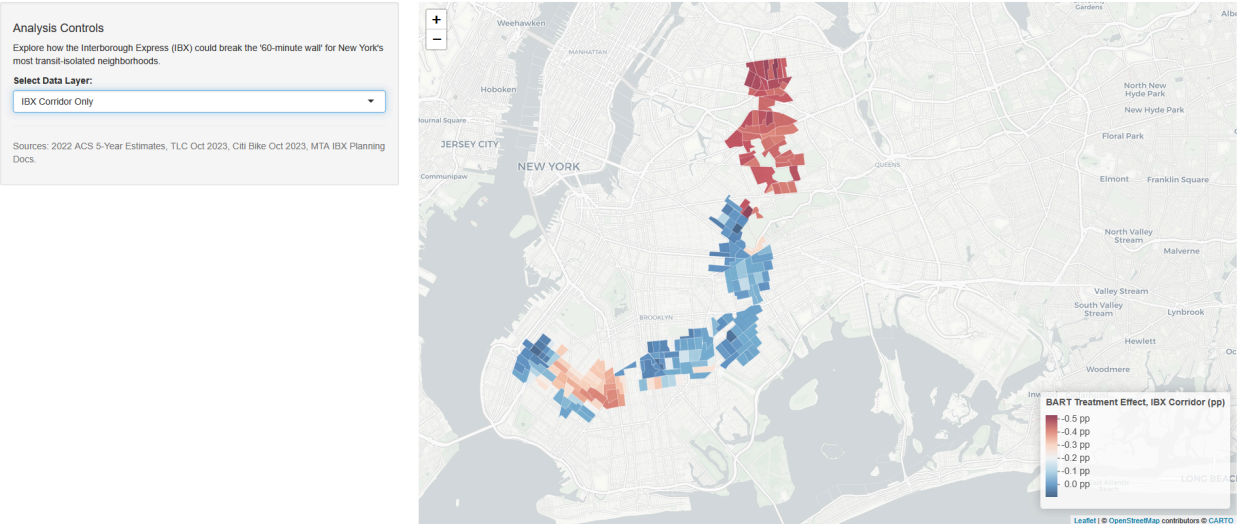


Figure 11. SHINY APP: BART TREATMENT MAP (IBX TRACTS ONLY)

## 5. Conclusions and Reflection

The core finding is that model choice matters a lot in a sensitivity analysis like this. The linear baseline predicts essentially zero IBX impact, not because transit access is unrelated to long commutes, but because the relationship saturates in a way that a linear model cannot represent. The spline and RF both predict around 7 to 8 percentage points of relief for directly served tracts, which is a more plausible estimate than zero. Anyone reporting only the linear output for a policy audience would understate the actual impact in the data.

The second main lesson is that geographic structure needs to be addressed explicitly. Adding one distance-to-Manhattan variable halved the spatial clustering in RF residuals and improved RMSE across all three models. The model had been using borough fixed effects to capture both borough location and Manhattan proximity at the same time. Once we added an explicit distance feature, the two were separated and the coefficients became easier to interpret.

Building on that geographic-structure lesson, we added spatial-lag versions of our geographic features so each tract's model could see neighbor-weighted averages of subway distance, Manhattan distance, transit share, and a few others, not just the tract's own values. Every predictive model class improved by about 0.1 to 0.15 RMSE with the lag features, and the IBX simulation picked up benefits to roughly 113 tracts neighboring the corridor that are not directly within walking distance of any station. Together with the 199 directly served tracts, our best model predicts measurable relief for tracts holding close to 1 million NYC residents, around 11% of the city's population.

### Limitations:

There are a few limitations worth noting. The simulation overstates benefits for tracts that already have decent transit access, since it sets subway distance to the best value in the dataset for all treated tracts. The target variable only covers workers who commute to work, missing non-work trips, remote workers, and people who may have already chosen shorter-commute jobs around transit gaps. And since the model is trained on a single snapshot of data, it cannot account for how patterns might shift over time.

### What's Next:

Future directions include validating against a real completed expansion (the Second Avenue Q extension is the obvious candidate), incorporating bus network coverage and job accessibility, and eventually moving toward a GTFS-based routing simulation for origin-destination pair estimates rather than tract-level aggregates.

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